

Operating Systems

2520030042

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SEC-5

WEEK-6

6. Create a client-server application using Named Pipes (FIFOs). The client sends messages and the server processes and responds to them. Analyze FIFO behavior when multiple clients communicate simultaneously.

Create a POSIX signal handling program that captures SIGINT, SIGTERM, and SIGUSR1. Demonstrate asynchronous event handling and explain the role of signal handlers.

Code

```
Ashish@Ashish: ~
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <signal.h>

void h(int s)
{
    printf("Signal %d received\n", s);
}

int main()
{
    char b[50];

    mkfifo("myfifo", 0666);

    signal(SIGINT, h);
    signal(SIGTERM, h);
    signal(SIGUSR1, h);

    printf("PID: %d\n", getpid());
    printf("Waiting for data...\n");

    int f = open("myfifo", O_RDONLY);
    read(f, b, 50);
    printf("Received: %s\n", b);
    close(f);

    printf("Waiting for signals...\n");

    while(1) sleep(1);

    return 0;
}
```

output

```
Ashish@Ashish: ~  
Ashish@Ashish:~$ nano signal.c  
Ashish@Ashish:~$ gcc signal.c  
Ashish@Ashish:~$ ./a.out  
PID: 481  
Waiting for data...  
^CSignal 2 received  
Received: Hello  
  
Waiting for signals...  
Signal 10 received|
```